

Analyzing and Predicting Amazon Customer Behavior

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Problem & Stakeholder

- We wish to examine customer behavior when it comes to e-commerce websites, mainly Amazon
- We want to use current Amazon customer activity to predict future Amazon customer activity
- Will further strengthen Amazon's business model by enhancing recommendations and increasing engagement with recommendations

Primary Stakeholder: Amazon Executives

- Increase sales, engagement, and conversion rates
- Internal scoring system of item popularity for Amazon to focus on which products to advertise and prioritize
- Increased buying patterns due to associated purchases, increasing overall “shopping cart” size
- An increased push towards products people tend to gravitate towards will generate more favorable opinions towards the company
- Further tailor and customize Amazon to the specific user profile

Secondary Stakeholder: Third-Party Amazon Sellers

- Small Businesses determine cross-category purchase patterns that reveal adjacent verticals
 - Ex. A candy buyer (current businesses) will also buys chocolate (new vertical)
- Sellers can use these patterns to identify complementary product lines worth expanding into
- Helps small sellers compete by data-driven product diversification

Envisioned Solution

Build a comprehensive product recommendation system that uses a combination of purchase histories and demographic factors, in addition to overall consumer purchase behavior, to predict future purchases and create appropriate recommendations.

Providing the most accurate recommendations possible is key to driving purchase activity.

This involves:

- Predicting consumers' future purchases based on past purchases
- Examining the impact of demographic features on purchase behavior
- Extrapolating our system beyond our sample data, which requires being able to deduce consumer demographics when not provided

Data

- 5 years worth of crowdsource Amazon purchase histories and their user demographics, spanning from 2018 to 2022
- Variables relating to user demographics were collected through an online survey
- Also includes details about the questions used in the survey and the format they were presented in
- Exported by Amazon users from the Amazon website and were shared by consenting users
- Multiple CSV files were included in this dataset

amazon-purchases.csv

- Details about the Amazon orders themselves, including the order date, state the shipping address is located in, purchase price, quantity of product purchased, product name, and category product can be found in among other values
- Also includes a column named SurveyResponsesID (randomly generated at the time of collection) which links the user's survey response to their Amazon purchase

survey.csv

- Survey responses including only responses from users who willingly chose to participate and share their data
- Also includes the SurveyResponseID column as a link with amazon-purchases.csv

fields.csv

- Names and descriptions of columns in survey.csv
- Fields/survey columns correspond to survey questions

Initial Results

Model	Test Accuracy	Improvement from Baseline
Random baseline (1/1,625)	0.06%	1×
KNN (no dim. reduction)	3.4%	57×
Random Forest (10K subsample)	7.28% (Azure)	121×
MLP	6.9%	115×
Wide & Deep + temporal features	7.04%	117×
LSTM	7.08%	117×
XGBoost	<i>Testing</i>	<i>Testing</i>

Initial Results

- Conducted sequential feature engineering to add variables for purchase order and time between purchases to enhance predictions
- **Our highest accuracy reached 7.28%**
- We built a range of neural networks, testing various hyperparameters
 - **K-Nearest Neighbors**
 - Tests whether demographically similar customers buy similar categories without any learned structure
 - **Random Forest**
 - Handles mixed feature types well, robust, and gives feature importance for stakeholders
 - **Multilayer Perceptron (MLP)**
 - Standard deep learning baseline
 - **Wide & Deep Neural Network**
 - Designed for recommendation problems: the wide branch memorizes feature crosses, the deep branch generalizes
 - Added temporal features (purchase order, days since last purchase)
 - **Long Short-Term Memory (LSTM)**
 - Tests whether the order of past purchases carried additional predictive value
 - **XGBoost**
 - Gradient-boosted trees, typically the strongest tabular baseline
 - Tuned via RandomizedSearchCV

Challenges

- **Memory Issues**
 - We mitigated this by obtaining an Azure instance for more computationally expensive
- **Unclear Baseline**
 - There is minimal information available as to the accuracy Amazon currently achieves with their predictive models, making it unclear what threshold we are aiming to improve on
- **Extreme Cardinality**
 - 1,600+ unique product categories
 - Difficult to group them together into fewer categories in an efficient manner
- **Data Leakage**
 - Data leakage bug discovered and fixed
StandardScaler and PCA were doing fit_transform on test data caught at midterm checkpoint and addressed

Plan & Goals

- Conduct Category Consolidation
 - 1,625 categories to 15-30 parent categories
 - Reduces memory for Random Forest/Tree-based models on the full dataset
 - No more subsampling
 - Decrease imbalance problem as class shrinks from 1,625 to 30
- Attempt a collaborative filtering model, similar to models that Amazon currently uses, and monitor performance
- Determine if we can identify user demographics from purchase behavior
 - This would allow us to extrapolate our model beyond just our current dataset, to typical Amazon users who do not provide demographic information
- Continue Testing on Azure Instance

Thank you!